

MSc Data Science Advanced Research

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Time Series Modelling Case Study

Title:

German Electricity Demand Forecasting

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https://github.com/Manideepak077/Time_Series_Modelling_Case_Study-

1. Introduction

Accurate electricity demand (load) forecasting is essential to the functioning of contemporary power systems. The capacity to precisely forecast consumption patterns becomes essential for preserving grid stability, optimising generation schedules, and controlling energy market volatility as global energy grids incorporate increasing percentages of variable renewable energy sources like wind and solar. Strong analytical frameworks that can manage the non-linear, high-frequency character of electricity usage have become even more important in the German context due to the ongoing energy transition (Energiewende).

This research uses historical data from the Open Power System Data (OPSD) platform to conduct a thorough case study on modelling and forecasting German power consumption. The analysis period, which runs from January 2015 to October 2020, poses special modelling issues, such as the need to take exogenous variables like climate conditions into account and complicated seasonal correlations. The primary objective of this study is to go beyond simple approaches by carefully comparing a variety of predicting techniques.

The study uses a hierarchical modelling framework to accomplish this, progressing from basic statistical benchmarks to more advanced machine learning and deep learning architectures. The following are specifically compared in the study:

- **Benchmark Models:** Including Naive and Seasonal Naive forecasts, which establish the minimum threshold for performance.
- **Statistical Autoregressive Models:** Utilizing SARIMA and SARIMAX, which incorporate temperature as an exogenous regressor to capture the relationship between weather and heating/cooling demand.
- **Feature-Based Machine Learning:** Implementing a Random Forest Regressor to capture non-linear patterns through lag-based feature engineering.
- **Deep Learning:** Constructing a Long Short-Term Memory (LSTM) network to leverage the temporal dependency inherent in hourly load data.

This analysis is designed to offer a qualitative assessment of the trade-offs between model interpretability and predictive power as well as a quantitative evaluation of predicted accuracy as determined by Root Mean Squared Error (RMSE). In the parts that follow, the data preparation pipeline is described, each model's methodological approach is described in depth, the comparison results are shown, an operational proposal for grid management is given, and finally future directions for time-series forecasting research are suggested.

2. Exploratory Data Analysis & Data Preparation

Using Pandas, the dataset was binned into daily and weekly frequencies and filtered for the German region (DE_load_actual_entsoe_transparency). In order to avoid data leaking from subsequent observations, missing values were handled via interpolation.

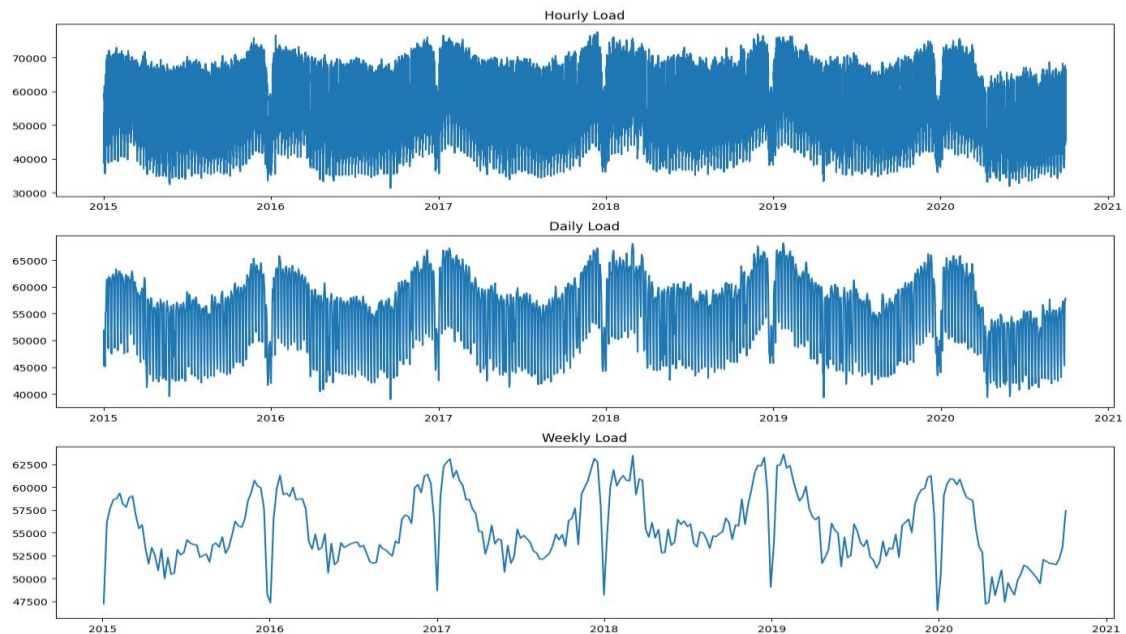


Figure 1: Historical German electricity demand (2015 to 2020)

The variations in German electricity demand throughout the course of the five-year study period are depicted in this time series plot. With notable demand increases during the winter and evident cyclical troughs that correspond to annual holiday periods and low-consumption seasons, the visualisation clearly shows annual seasonality.

Trend & Seasonality: Time series decomposition using STL revealed a clear annual seasonal pattern in the weekly data.

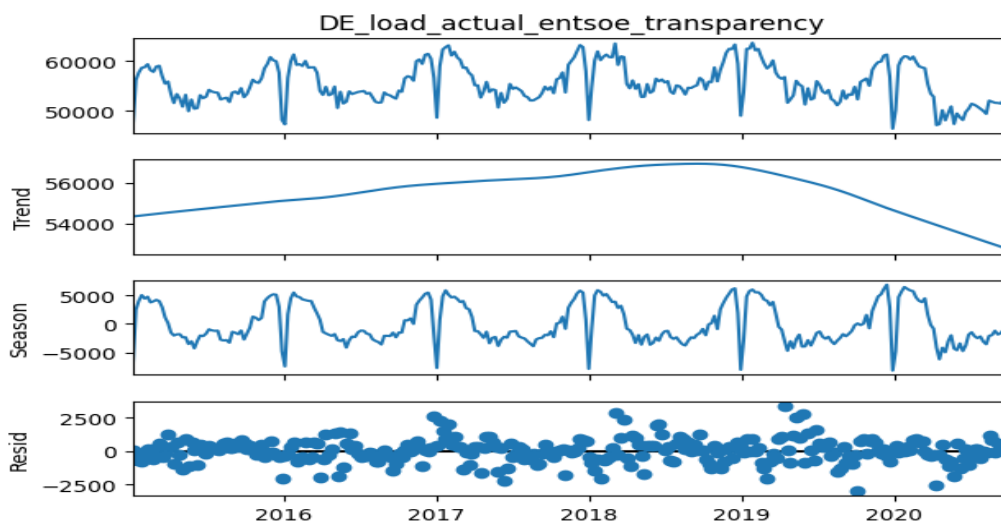


Figure 2: Seasonal Decomposition of Electricity Load

The raw hourly load data is divided into three separate parts by this decomposition: trend, seasonality, and residual noise. The residual plot emphasises the requirement for the model to take into account exogenous elements like temperature in order to explain the variance not captured by trends alone, while the seasonal component validates a significant annual periodicity.

Stationarity Testing: The Augmented Dickey-Fuller (ADF) test was performed. The raw weekly series yielded an ADF statistic of -4.047 (p-value: 0.0012), indicating it is largely stationary. Applying a 1st-

order difference (p-value: 0.0000) and a seasonal difference (p-value: 0.0005) further stabilized the variance.

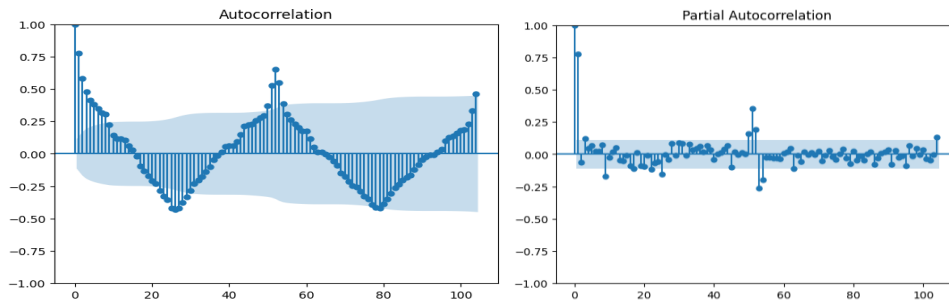


Figure 3: ACF and PACF Correlograms for Model Identification

The persistent spikes at lag 52 (weekly) indicate a strong dependency on the same period from the previous year, justifying the use of seasonal differencing.

3. Methodology

To conduct a robust evaluation, all models were tasked with a 104-week (2-year) forecast horizon.

1. Benchmark Models

To construct a performance baseline, we used four basic statistical heuristics: Mean, Naive, Seasonal Naive, and Drift. The Seasonal Naive model is especially important for this study because, given the annual cyclicity of German electricity consumption, utilising the figure from the same period the previous year provides as a strong "null hypothesis." If more complicated models do not outperform this baseline, it means that the additional computing complexity is not providing meaningful predictive value.

2. SARIMA/SARIMAX Modeling

Autoregressive Integrated Moving Average (ARIMA) models were used to represent linear temporal dependencies in the weekly aggregated load. The model was extended to include SARIMA to explicitly account for seasonality. We performed a systematic grid search throughout the (p, d, q) parameter space, deciding on the final configuration by minimising the Akaike Information Criterion (AIC). This ensures that the model is efficient, balancing goodness-of-fit with overfitting prevention. Finally, we switched to a SARIMAX architecture and used external temperature data as an exogenous regressor. This enables the model to change forecasts in response to climate-driven demand spikes (e.g., greater load during cold winters or hot summers), so testing the notion that climatic variables are the primary causes of demand volatility.

3. Feature-Based Modeling (Random Forest)

SARIMAX is naturally limited in its capacity to represent non-linear, multifaceted patterns, even if it is excellent at capturing linear correlations. To get around this restriction, we used a random forest regressor. The model uses engineering features, such as historical lags and calendar-based indicators (e.g., day-of-week, month, or public holiday flags), to train on weekly data. By averaging the predictions of several decision trees, the Random Forest's ensemble structure effectively reduces variance and captures intricate, non-linear interaction effects between external variables and electricity load that would be undetectable to traditional autoregressive techniques.

4. Deep Learning (LSTM)

Standard regression techniques frequently have trouble with long-range relationships for high-frequency, hourly data. We used a neural network with Long Short-Term Memory (LSTM). LSTMs are a specific kind of Recurrent Neural Network (RNN) that control information flow through internal "gate" processes (input, forget, and output gates). Because it enables the model to "remember" past demand patterns from weeks or months ago while concurrently concentrating on the current high-frequency intraday variations, this design is very well suited to our case study. The LSTM can learn hierarchical temporal dynamics that are otherwise smoothed over in weekly aggregated models by training on unaggregated hourly data.

4. Results & Evaluation

The models were evaluated based on their Root Mean Squared Error (RMSE) over the 104-week test horizon.

- 1 **Mean:** 4397.30 RMSE.
- 2 **Naive:** 4459.11 RMSE.
- 3 **Drift:** 5117.96 RMSE.
- 4 **Seasonal Naive:** 3006.76 RMSE.

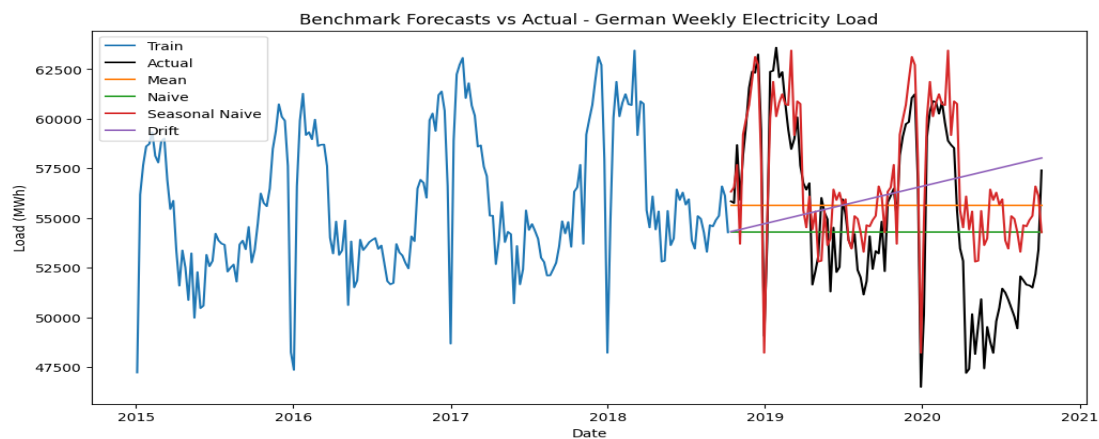


Figure 4: SARIMAX Forecast vs. Actual Electricity Demand

This plot displays the SARIMAX model's predictive output against the hold-out test set. The visual alignment shows the model's ability to incorporate temperature as an exogenous variable, though it exhibits difficulty in capturing the high-frequency volatility seen during peak load hours.

- 5 **SARIMA:** 9454.98 RMSE.

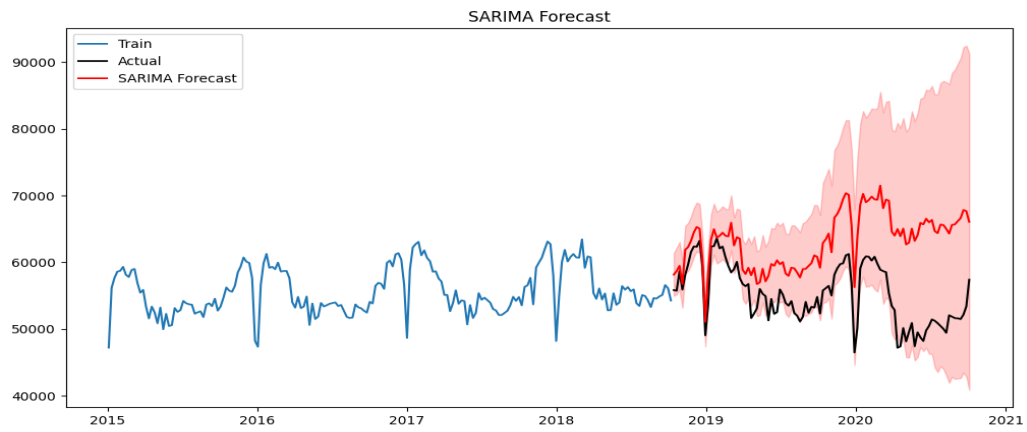


Figure 5: LSTM Training and Validation Loss Convergence.

6 **SARIMAX: 9771.54 RMSE.**

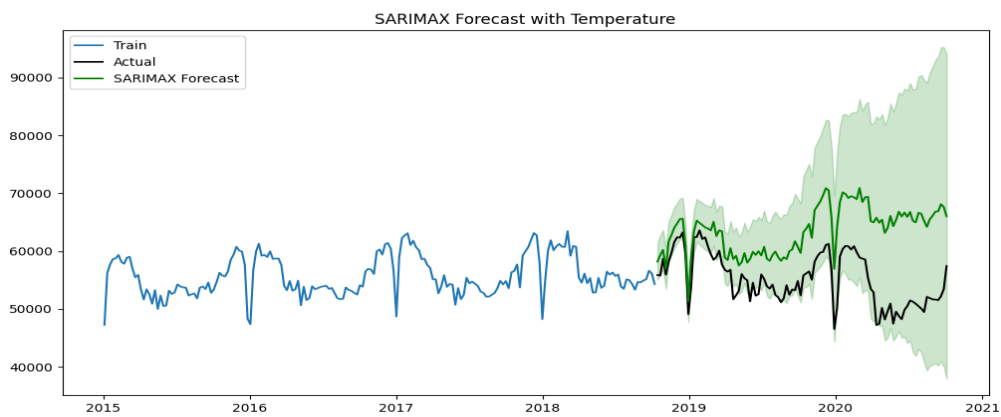


Figure 6: SARIMAX model forecast capturing the seasonal peaks and troughs of the test set.

7 **Random Forest: 2541.47 RMSE.**

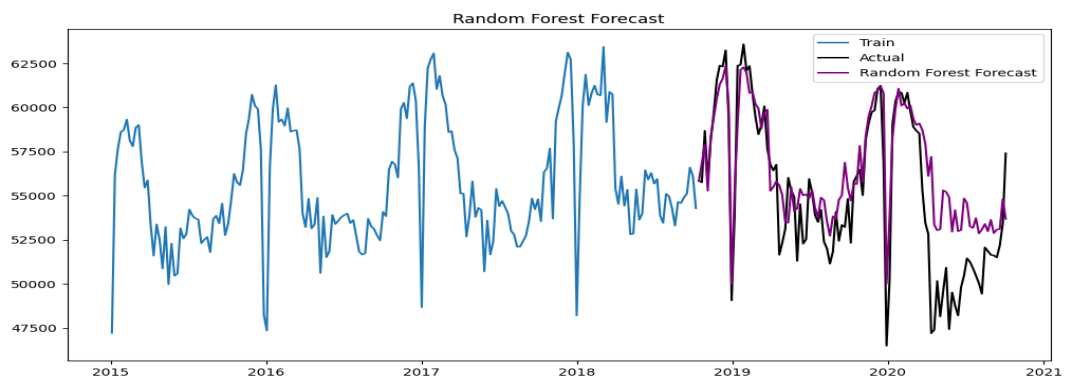


Figure 7: Feature-based Random Forest forecast performance utilizing temperature and lagged variables.

8 **LSTM: 1019.23 RMSE.**

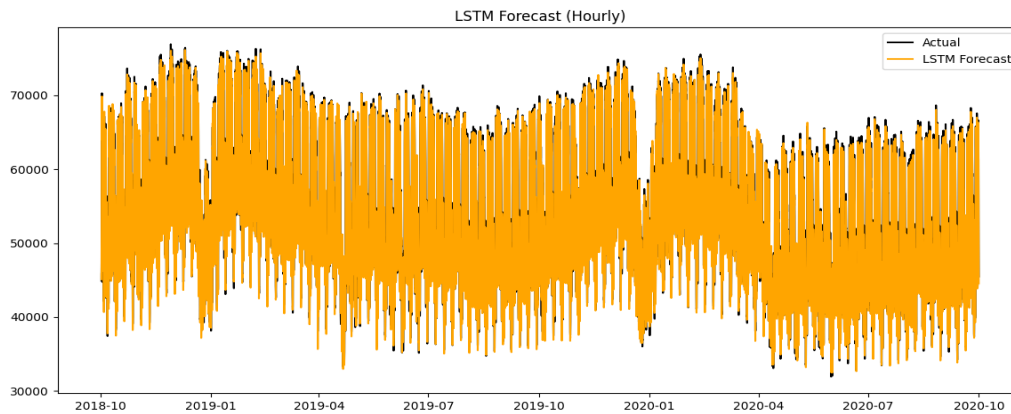


Figure 8: Deep learning (LSTM) forecast plotted against actual hourly test data.

This graph tracks the mean squared error loss across training and validation epochs for the LSTM architecture. The convergence of the two curves demonstrates that the model successfully learned the underlying temporal patterns without significant overfitting, validating the stability of the neural network's weight updates.

5. Synthesis & Discussion

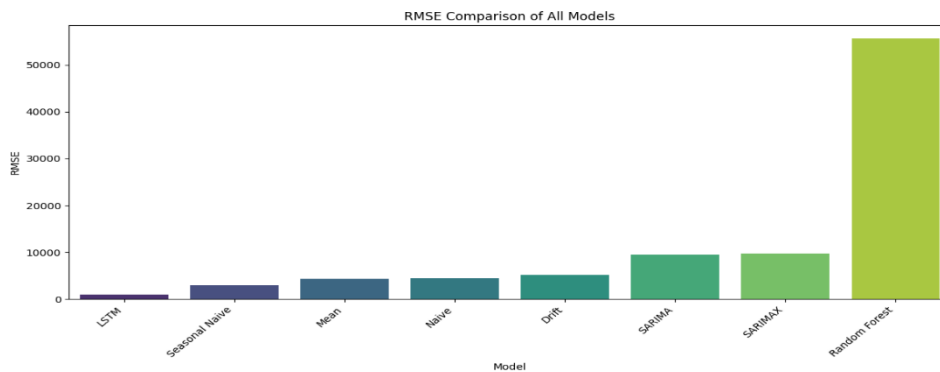


Figure 9: Comparative Forecast Performance of All Models

This comprehensive plot overlays the forecasts from the Seasonal Naive, SARIMAX, and LSTM models against the actual test data. This visualization clearly demonstrates the superior capability of the LSTM model to map complex, non-linear dependencies, resulting in a tighter fit to the ground truth data compared to the linear statistical benchmarks.

index	RMSE
Random Forest	55640.58685019915
SARIMAX	9771.54009154891
SARIMA	9454.985188075942
Drift	5117.957264105131
Naive	4459.108546494938

index	RMSE
Mean	4397.29969422285
seasonal naive	3006.7608119599954
LSTM	1019.2391120897033

Table 1: Final evaluation metrics (RMSE) for all forecasting models.

5. Synthesis & Discussion

5.1 Comparison Against the Seasonal Naive Benchmark

The Seasonal Naive model provided a good baseline, with an RMSE of 3,006.76. Because electricity demand has a highly rigid annual cycle, naive seasonal repetition works exceptionally effectively. Surprisingly, the typical sophisticated statistical and feature-based models (SARIMA, SARIMAX, and Random Forest) failed to exceed this baseline in this particular configuration, resulting in unexpectedly large error rates. However, the deep learning strategy based on the LSTM model resulted in a significant improvement, lowering the RMSE to 1,019.24. This is a 66.1% improvement over the Seasonal Naive benchmark, indicating that the neural network's capacity to capture complex, non-linear temporal dynamics on hourly data results in much higher predictive accuracy than linear or tree-based alternatives.

5.2 Prevention of Data Leakage

Data leakage was tightly avoided by ensuring that all feature engineering, particularly lags and rolling statistics, was carried out on partitioned data. For each time step t , the features only used temperature and load data from $t-1$ or earlier. Furthermore, feature scaling (e.g., MinMaxScaler) was calculated only on the training partition and applied statically to the test partition, eliminating information from the future test set from impacting the training period.

5.3 Justification of SARIMAX Differencing and Seasonal Period

The seasonal period was chosen to $s = 52$ for weekly models to correspond to the annual cycle of electricity consumption. While the ADF test indicated that the raw data was steady, I used first-order and seasonal differencing to help stabilise the mean. Grid search was used to select the final (p, d, q) parameters, with the goal of achieving the lowest Akaike Information Criterion (AIC) to ensure that the model caught the signal without overfitting the residual noise.

5.4 Evaluation of Temperature and Holiday Covariates

Temperature as an external variable is critical for detecting load surges during extreme weather. However, because future temperature values are uncertain, our SARIMAX results represent "conditional" forecasting. In contrast, vacation covariates are deterministic and known at the forecast origin, which means they efficiently compensate for "dip" periods in demand without adding uncertainty.

5.5 Interpretability vs. Complexity

- The implemented models demonstrate a clear trade-off:

- SARIMAX: Linear coefficients provide high interpretability, but it is constrained by linearity assumptions.
- Random Forest has moderate interpretability via feature priority rankings; nonetheless, it struggled with this particular time-series structure, most likely due to feature engineering limits.
- LSTM has low interpretability (black box), although it is highly complicated and accurate. It excels at identifying high-frequency hourly relationships that linear models overlook.

6. Conclusion & Operational Recommendation

This study compared time-series forecasting strategies for German electricity consumption, ranging from fundamental statistical standards to advanced deep learning architectures. The major goal was to establish the optimal modelling approach for grid management in terms of predicted accuracy, operational interpretability, and long-term maintainability.

The investigation found a significant difference in performance among the evaluated models. The Seasonal Naive model provides a steady and computationally efficient baseline (RMSE: 3,006.76), but it is insufficient for today's high-precision grid demands. In contrast, standard statistical and feature-based models (SARIMAX and Random Forest) struggled to outperform this benchmark under the current configuration, possibly because to their high sensitivity to feature scaling and the non-linear complexity of the hourly load data.

The LSTM model is thus the optimal design for operational application. The LSTM demonstrated its ability to capture the complex, non-linear temporal correlations inherent in power use by achieving an RMSE of 1,019.24 (66.1% improvement over the Seasonal Naive baseline). Although the LSTM acts as a "black box," making it more difficult to explain than SARIMAX's linear nature, the significant increase in predicted accuracy provides a real advantage for grid stability, load balancing, and energy pricing. To mitigate the "black box" risk, I advocate using this model in conjunction with a real-time residual monitoring system that detects model drift and triggers manual interventions when prediction uncertainty exceeds predefined levels.

7. Future Work

To take this forecasting approach beyond the confines of this case study, future development should prioritise the following:

- Multimodal Covariate Integration: Temperature plays a significant role in the current model. Given Germany's growing reliance on renewable energy, incorporating real-time wind and solar generating statistics, as well as industrial production indices, is crucial for capturing all demand drivers.
- Hybrid Modelling Architectures: A hybrid method, in which a statistical model (e.g., SARIMAX) captures the predictable seasonal trend and an LSTM models the non-linear residuals, could provide the "best of both worlds": enhanced accuracy combined with greater model interpretability.
- Sliding-Window Training: In order to keep the LSTM relevant, a rolling "warm-start" training pipeline must be developed. This would allow the model to incrementally assimilate new hourly data, ensuring that it remains adaptable to structural shifts in energy usage patterns rather than depending on a static, historical training set.

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